

PUNJABI UNIVERSITY CENTRE FOR
EMERGING AND INNOVATIVE TECHNOLOGY, MOHALI

MST II-BEE
MAX.MARKS-15

B. TECH (CSE)-1st (SEM-I)
TIME ALLOWED-1 HR

Note: Attempt any three questions. All questions carry equal marks.

Q.1 Solve the followings:

a) Subtract $(35)_{10}$ from $(18)_{10}$ using the *Complementary* method.

(2.5)

b) Convert $(11010011)_2 = (?)_{16}$

(2.5)

Q.2 State and prove the two basic De Morgan's theorem.

(5)

Q.3 Identify the basic elements of a communication system and the purpose of each in detail.

(5)

Q.4 Write a short note on the followings:

(5)

a.) Excess 3 code

b.) NAND gate

c.) EX-OR gate

CLASS: -

PUNJABI UNIVERSITY, PATIALA

B. Tech (CSE) 1st Year 1st sem MST-2 Max. Marks = 15

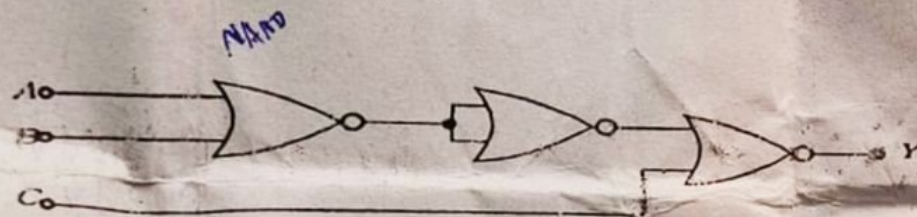
SUBJECT: - Basic Electronics Engineering (ECEB1123T) Date: - Nov. 2024 TIME: - 1 Hr

Note: - Section A is compulsory and attempt any 2 questions from section B

Q 1

Section A

- Draw and write truth table of universal gates.
- Why there is need of modulation?
- Write the truth table of a 2-input EX-NOR gate?
- Find 2's complement of number 1110011?
- Solve the network in the given figure for Y:



Section B

- Explain in detail the AM Radio transmitter with the help of block diagram?
- a) State and prove De-Morgan's first and second Theorem?
b) Encode the decimal number 14 to Gray code
c) Convert decimal 46 into Excess-3 code.
- Simplify the following Boolean Algebra.

a) $\bar{A}B + A\bar{B}C + AB\bar{C} + ABC$

$$\bar{A}(A + BC) + (AC + \bar{B}C)$$

2.5

2.5

32 16 8 4 2 1
1 0 1 1 1 0
32 16 8 4 2 1
1 0 1 1 1 0

(1)
(1)
(1)
(1)

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$
$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

(1)

(5)

(3)

(1)

(1)

(1110)
1001

32 16 8 4 2 1
0 0 0 0 0 0
0 0 0 1 0 0
0 0 1 0 0 0
0 0 0 1 0 0
0 0 0 1 0 0
0 0 0 1 0 0

Ben 421

NAND
A+B

A+B

A+B